# ECON 2050 – Mathematical Economics

## Convexity

# Convex combinations and line segments

## Definition

A **convex combination** of two vectors $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$ is any vector $\boldsymbol{z} \in \mathbb{R}^n$ such that

$$\boldsymbol{z} = \lambda \boldsymbol{x} + (1 - \lambda)\boldsymbol{y}, \quad \text{for some } \lambda \text{ such that } 0 \leq \lambda \leq 1.$$

## Definition

The **line segment** $[\boldsymbol{x}, \boldsymbol{y}]$ between $\boldsymbol{x} \in \mathbb{R}^n$ and $\boldsymbol{y} \in \mathbb{R}^n$ is the set of all convex combinations of $\boldsymbol{x}$ and $\boldsymbol{y}$. That is,

$$[\boldsymbol{x}, \boldsymbol{y}] = \{\boldsymbol{z} \in \mathbb{R}^n : \boldsymbol{z} = \lambda \boldsymbol{x} + (1 - \lambda)\boldsymbol{y} \text{ for some } \lambda \text{ such that } 0 \leq \lambda \leq 1\}$$

Examples:

- $[0,1]$ = line segment between 0 and 1 = closed interval between 0 and 1
- $[(1,0),(3,4)] = \{(x,y) : (x,y) = \lambda(1,0) + (1-\lambda)(3,4),\ 0 \leq \lambda \leq 1\}$
  $=$ line segment between $(1,0)$ and $(3,4)$.

  If $\lambda = 0$ we are at $(3,4)$; if $\lambda = 1$ we are at $(1,0)$; if $\lambda = \frac{1}{2}$ we are at the midpoint $(2,2) = \frac{1}{2}(1,0) + \frac{1}{2}(3,4)$ of the segment.

Note that the set

$$\{\boldsymbol{z} \in \mathbb{R}^n : \boldsymbol{z} = \lambda \boldsymbol{x} + (1-\lambda)\boldsymbol{y} \text{ for some } \lambda \in \mathbb{R}\}$$
$$= \{\boldsymbol{z} \in \mathbb{R}^n : \boldsymbol{z} = \boldsymbol{y} + \lambda(\boldsymbol{x} - \boldsymbol{y}) \text{ for some } \lambda \in \mathbb{R}\}$$

describes the straight line through the points $\boldsymbol{x}$, $\boldsymbol{y}$.

# Convex sets

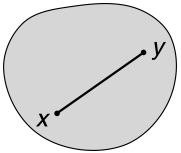
## Definition

A set $S \subseteq \mathbb{R}^n$ is **convex** if for every two points of the set the line segment between the two points belongs to the set, that is, if

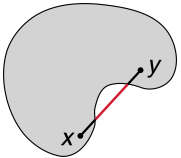
$$\forall \boldsymbol{x}, \boldsymbol{y} \in S : [\boldsymbol{x}, \boldsymbol{y}] \subseteq S,$$

or, equivalently, if

$$\forall \boldsymbol{x}, \boldsymbol{y} \in S, \forall \lambda \in [0,1] : \lambda \boldsymbol{x} + (1-\lambda)\boldsymbol{y} \in S$$



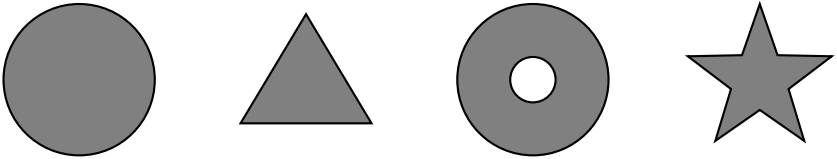

A convex set




A set that is not convex

Examples: Intervals and (more generally) balls are convex sets.

Exercise: Determine whether the following four sets (in grey) are convex.



convex convex not convex not convex

Example: $T = \{(x, y) \in \mathbb{R}^2 : x + 2y \leq 3\}$ is convex: Let $(x', y'), (x'', y'') \in T$ and $\lambda \in [0, 1]$. We need to show that

$$\lambda(x', y') + (1 - \lambda)(x'', y'') = (\lambda x' + (1 - \lambda)x'', \lambda y' + (1 - \lambda)y'') \in T,$$

or, equivalently, that

$$(\lambda x' + (1 - \lambda)x'') + 2(\lambda y' + (1 - \lambda)y'') \leq 3. \tag{1}$$

But $(x', y') \in T$ implies $x' + 2y' \leq 3$ and thus

$$\lambda(x' + 2y') \leq 3\lambda, \tag{2}$$

and $(x'', y'') \in T$ implies $x'' + 2y'' \leq 3$ and thus

$$(1 - \lambda)(x'' + 2y'') \leq 3(1 - \lambda). \tag{3}$$

Adding up the two inequalities (2) and (3), we obtain (1).

**Proposition**

*Every intersection of convex sets is a convex set.*

Proof:

- Suppose $S_i$ for $i \in \mathcal{I}$, are convex sets (where $\mathcal{I}$ is some "index" set, e.g. $\mathcal{I} = \{1, 2, 3\}$).
- If $\boldsymbol{x}, \boldsymbol{y} \in \bigcap_{i \in \mathcal{I}} S_i$ then $\boldsymbol{x}, \boldsymbol{y} \in S_i$ for every $i \in \mathcal{I}$.
- For every $i$, since $S_i$ is convex, the line segment between $\boldsymbol{x}$ and $\boldsymbol{y}$, $[\boldsymbol{x}, \boldsymbol{y}]$, is contained in $S_i$.
- Therefore, $[\boldsymbol{x}, \boldsymbol{y}]$ belongs to the intersection $\bigcap_{i \in \mathcal{I}} S_i$.

Example: $S = \left\{(x, y) \in \mathbb{R}^2 : x + 2y \leq 3,\ x \geq 0,\ y \geq 0\right\}$ is convex because

$$S = T \cap \{(x, y) \in \mathbb{R}^2 : x \geq 0\} \cap \{(x, y) \in \mathbb{R}^2 : y \geq 0\},$$

where $T = \left\{(x, y) \in \mathbb{R}^2 : x + 2y \leq 3\right\}$, and it is not difficult to check that both $\{(x, y) \in \mathbb{R}^2 : x \geq 0\}, \{(x, y) \in \mathbb{R}^2 : y \geq 0\}$ are convex (exercises; use definition).

Warning: In general, the union of convex sets is not a convex set.

Exercise: Write down or draw two convex sets $S$ and $T$ whose union $S \cup T$ is not convex.

- $S = \{1\}$ and $T = \{2\}$ are convex, but $S \cup T = \{1, 2\}$ is not convex.
- $S = [1, 2]$ and $T = [3, 4]$ are convex, but $S \cup T = [1, 2] \cup [3, 4]$ is not convex: $x' = 2$ and $x'' = 3$ are contained in $S \cup T$, but $(1/2)x' + (1/2)x'' = (1/2)2 + (1/2)3 = 2.5 \notin S \cup T$
- $S = B_1(1, 0)$ and $T = B_1(-1, 0)$ are convex but $S \cup T$ is not convex.

  (E.g. $(1, \frac{1}{2}) \in S$ and $(-1, \frac{1}{2}) \in T$ both belong to $S \cup T$, but the midpoint $(0, \frac{1}{2})$ of the line segment $[(1, \frac{1}{2}), (-1, \frac{1}{2})]$ does not belong to either of the two sets and, therefore, it does not belong to the union of them.)

# Convex and Concave Functions (SB Chapter 21.1)

## Definition

Let $S \subseteq \mathbb{R}^n$ be convex. A function $f : S \to \mathbb{R}$ is **convex** (on $S$) if $\forall \boldsymbol{x}, \boldsymbol{y} \in S$, $\forall \lambda \in [0,1]$

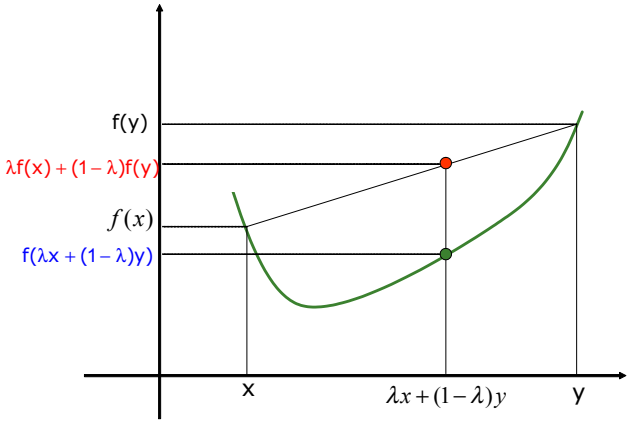
$$f(\lambda \boldsymbol{x} + (1-\lambda)\boldsymbol{y}) \leq \lambda f(\boldsymbol{x}) + (1-\lambda) f(\boldsymbol{y}).$$

The function is **strictly convex** (on $S$) if $\forall \boldsymbol{x}, \boldsymbol{y} \in S$ such that $\boldsymbol{x} \neq \boldsymbol{y}$, $\forall \lambda \in (0,1)$,

$$f(\lambda \boldsymbol{x} + (1-\lambda)\boldsymbol{y}) < \lambda f(\boldsymbol{x}) + (1-\lambda) f(\boldsymbol{y}).$$

A function is convex if for all $\boldsymbol{x}$ and $\boldsymbol{y}$, the graph of the function over the line segment between $\boldsymbol{x}$ and $\boldsymbol{y}$ lies below the line segment between the points $(\boldsymbol{x}, f(\boldsymbol{x}))$ and $(\boldsymbol{y}, f(\boldsymbol{y}))$.

A strictly convex function of one variable:



## Definition

Let $S \subseteq \mathbb{R}^n$ be convex. A function $f : S \to \mathbb{R}$ is **concave** (on $S$) if $\forall \boldsymbol{x}, \boldsymbol{y} \in S$, $\forall \lambda \in [0,1]$

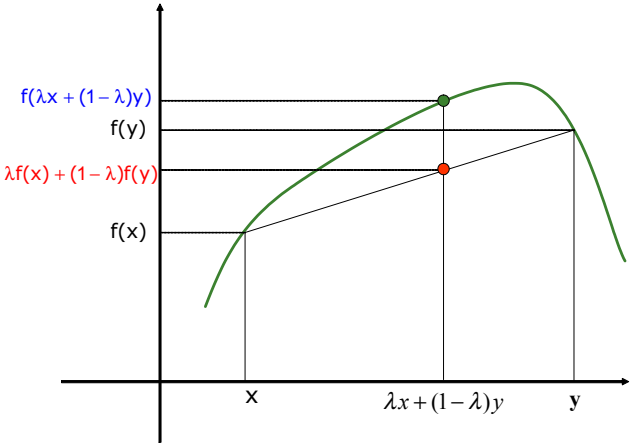
$$f(\lambda \boldsymbol{x} + (1-\lambda)\boldsymbol{y}) \geq \lambda f(\boldsymbol{x}) + (1-\lambda) f(\boldsymbol{y}).$$

The function is **strictly concave** (on $S$) if $\forall \boldsymbol{x}, \boldsymbol{y} \in S$ such that $\boldsymbol{x} \neq \boldsymbol{y}$, $\forall \lambda \in (0,1)$,

$$f(\lambda \boldsymbol{x} + (1-\lambda)\boldsymbol{y}) > \lambda f(\boldsymbol{x}) + (1-\lambda) f(\boldsymbol{y}).$$

A function is concave if for all $\boldsymbol{x}$ and $\boldsymbol{y}$, the graph of the function over the line segment between $\boldsymbol{x}$ and $\boldsymbol{y}$ lies above the line segment between the points $(\boldsymbol{x}, f(\boldsymbol{x}))$ and $(\boldsymbol{y}, f(\boldsymbol{y}))$.

A strictly concave function of one variable:



Examples:

- $f(x_1, x_2) = x_1 + x_2$ is both convex and concave:

  Let $(x_1', x_2'), (x_1'', x_2'') \in \mathbb{R}^2$ and $\lambda \in [0, 1]$, then

$$\begin{aligned} f(\lambda(x_1', x_2') + (1-\lambda)(x_1'', x_2'')) &= f(\lambda x_1' + (1-\lambda)x_1'', \lambda x_2' + (1-\lambda)x_2'') \\ &= \lambda x_1' + (1-\lambda)x_1'' + \lambda x_2' + (1-\lambda)x_2'' \\ &= \lambda(x_1' + x_2') + (1-\lambda)(x_1'' + x_2'') \\ &= \lambda f(x_1', x_2') + (1-\lambda) f(x_1'', x_2'') \end{aligned}$$

  Thus $f$ is both concave and convex, but neither strictly concave nor strictly convex.

  More generally, every affine function $f : \mathbb{R}^n \to \mathbb{R}$ is both convex and concave (exercise).

- $f_1(x) = x^2$ and $f_2(x) = e^x$ are strictly convex, $f_3(x) = \ln x$ and $f_4(x) = \sqrt{x}$ are strictly concave.

# Increasing and decreasing functions: A review

## Definition

Let $T$ be a nonempty subset of $\mathbb{R}$. A function $f : T \to \mathbb{R}$ is said to be

- **increasing** if for any $x, y \in T$, $x \geq y$ implies $f(x) \geq f(y)$;
- **strictly increasing** if for any $x, y \in T$, $x > y$ implies $f(x) > f(y)$;
- **decreasing** if for any $x, y \in T$, $x \geq y$ implies $f(x) \leq f(y)$; and
- **strictly decreasing** if for any $x, y \in T$, $x > y$ implies $f(x) < f(y)$.

- We call a function that is either increasing or decreasing **monotone**.
- It is not difficult to see that $f$ is (strictly) decreasing if and only if $-f$ is (strictly) increasing.

**Theorem**

*Let $I$ be an open interval and $f : I \to \mathbb{R}$ be differentiable, then*

(1) *$f$ is increasing if and only if $f' \geq 0$ on $I$;*

(2) *if $f' > 0$ on $I$, then $f$ is strictly increasing;*

(3) *$f$ is decreasing if and only if $f' \leq 0$; and*

(4) *if $f' < 0$ on $I$, then $f$ is strictly decreasing.*

Note that the converse of parts (2) and (4) need not hold: suppose $f(x) = x^3$, we have $f'(0) = 0$ but $f$ is strictly increasing.

# Differentiable convex/concave function of one variable: A review

## Theorem

*Let $I$ be an open interval and $f : I \to \mathbb{R}$ be differentiable, then*

- *$f$ is (strictly) convex if and only if $f'$ is (strictly) increasing;*
- *$f$ is (strictly) concave if and only if $f'$ is (strictly) decreasing.*

## Corollary

*Let $I$ be an open interval and $f : I \to \mathbb{R}$ be differentiable, then*

- *$f$ is convex if and only if $f'' \geq 0$, and if $f'' > 0$, $f$ is strictly convex;*
- *$f$ is concave if and only if $f'' \leq 0$, and if $f'' < 0$, $f$ is strictly concave.*

# Properties of convex/concave functions

**Theorem (SB Theorem 21.8)**

*Let $S \subseteq \mathbb{R}^n$ be convex, and $f_1, f_2, \ldots, f_k$ be functions from $S$ to $\mathbb{R}$. If $f_1, f_2, ..., f_k$ are convex [concave], then for all $\boldsymbol{\alpha} \in \mathbb{R}^k$ such that $\alpha_i \geq 0$ for all $i = 1, \ldots, k$,*

$$\alpha_1 f_1 + \alpha_2 f_2 + ... + \alpha_k f_k$$

*is also convex [concave].*

Proof: Follows directly from the definition of convex/concave functions. A good (finger) exercise.

**Theorem**

*Let $S \subseteq \mathbb{R}^n$ be convex, and $f$ is a function from $S$ to $\mathbb{R}$.*

1. *If $f$ is convex [concave] then $-f : S \to \mathbb{R}, \boldsymbol{x} \mapsto -f(\boldsymbol{x})$ is concave [convex].*
2. *If $f$ is convex [concave] and $g : \mathbb{R} \to \mathbb{R}$ is a convex [concave] and increasing function, then $g \circ f$ is convex [concave].*

Proof: Part 1 follows directly from the definition, because minus sign reverses inequalities; for part 2, just apply the definition twice and use the fact that $g$ is increasing to "pass" the inequality.

Warning: If $g$ is not increasing, part 2 need not be true. E.g.,

- $f(x) = x^2$, is convex; $g(y) = -y$ is convex but not increasing.
- But $g \circ f : \mathbb{R} \to \mathbb{R}$, given by $g(f(x)) = -x^2$ is not convex.

Examples:

- $f : \mathbb{R} \to \mathbb{R}, x \mapsto 3x^2 + 2x$ is convex because
  - $f_1 : \mathbb{R} \to \mathbb{R}, x \mapsto x^2$ is convex,
  - $f_2 : \mathbb{R} \to \mathbb{R}, x \mapsto x$ is convex, and
  - $f = 3f_1 + 2f_2$

- $g : \mathbb{R} \to \mathbb{R}, x \mapsto -x^2$ is concave because
  - $f_1 : \mathbb{R} \to \mathbb{R}, x \mapsto x^2$ is convex,
  - $g = -f_1$

- $h : \mathbb{R}^2_{++} \to \mathbb{R}, (x, y) \mapsto \sqrt{5x + 2y}$ is concave because
  - $f : \mathbb{R}^2_{++} \to \mathbb{R}_+, (x, y) \mapsto 5x + 2y$ is affine (and therefore concave),
  - $g : \mathbb{R}_+ \to \mathbb{R}, x \mapsto \sqrt{x}$ is concave and increasing, and
  - $h = g \circ f$

We defined when a function is convex, and when a set is convex.

- To distinguish these concepts, we could say "function-convex" and "set-convex," but commonly both concepts are simply called "convex."
- The next proposition shows how "function-convexity" and "set-convexity" are related.

**Definition**

Let $S \subseteq \mathbb{R}^n$ be convex, and $f : S \to \mathbb{R}$. The **epigraph** of $f$ is the set

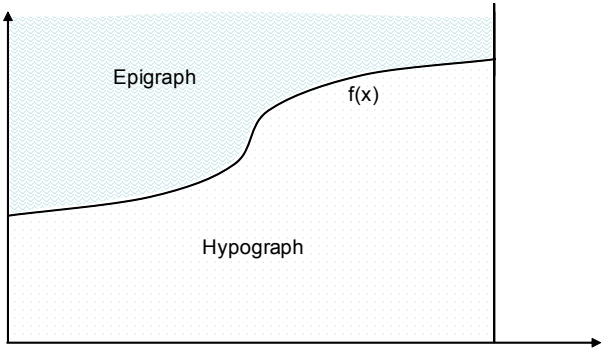
$$\text{epi}(f) = \{(\boldsymbol{x}, y) \in S \times \mathbb{R} : y \geq f(\boldsymbol{x})\},$$

and the **hypograph** of $f$ is the set

$$\text{hyp}(f) = \{(\boldsymbol{x}, y) \in S \times \mathbb{R} : y \leq f(\boldsymbol{x})\}.$$

Intuitively, the epigraph (hypograph) of a function is the set of all points that lie on or above (below) its graph.

Epigraph and hypograph of a function of one variable:



## Proposition

*Let $S \subseteq \mathbb{R}^n$ be convex. A function $f : S \to \mathbb{R}$ is*

- *concave if and only if its hypograph is a convex set.*
- *convex if and only if its epigraph is a convex set.*

Proof (first part): "⇒": Suppose that $f$ is concave. We want to show that $\text{hyp}(f)$ is a convex set.

Let $(\boldsymbol{x}_1, y_1), (\boldsymbol{x}_2, y_2) \in \text{hyp}(f)$, and $\lambda \in [0, 1]$. We have to show that $\lambda(\boldsymbol{x}_1, y_1) + (1 - \lambda)(\boldsymbol{x}_2, y_2) \in \text{hyp}(f)$.

Since $(\boldsymbol{x}_1, y_1) \in \text{hyp}(f)$, $y_1 \leq f(\boldsymbol{x}_1)$ and thus

$$\lambda y_1 \leq \lambda f(\boldsymbol{x}_1) \tag{4}$$

Since $(\boldsymbol{x}_2, y_2) \in \text{hyp}(f)$, $y_2 \leq f(\boldsymbol{x}_2)$ and thus

$$(1 - \lambda) y_2 \leq (1 - \lambda) f(\boldsymbol{x}_2) \tag{5}$$

Thus

$$\lambda y_1 + (1-\lambda)y_2 \leq \lambda f(\boldsymbol{x}_1) + (1-\lambda)f(\boldsymbol{x}_2) \leq f(\lambda \boldsymbol{x}_1 + (1-\lambda)\boldsymbol{x}_2),$$

where the first inequality follows from adding up (4) and (5) and the second inequality holds because $f$ is concave. We conclude that

$$(\lambda \boldsymbol{x}_1 + (1-\lambda)\boldsymbol{x}_2, \lambda y_1 + (1-\lambda)y_2) = \lambda(\boldsymbol{x}_1, y_1) + (1-\lambda)(\boldsymbol{x}_2, y_2) \in \text{hyp}(f)$$

Therefore, $\text{hyp}(f)$ is convex.

"$\Leftarrow$": Now suppose that $\text{hyp}(f)$ is convex. We want to show that $f$ is concave.

Take $\boldsymbol{x}_1, \boldsymbol{x}_2 \in S$ and let $\lambda \in [0,1]$. We have to show that

$$\lambda f(\boldsymbol{x}_1) + (1-\lambda) f(\boldsymbol{x}_2) \leq f(\lambda \boldsymbol{x}_1 + (1-\lambda)\boldsymbol{x}_2).$$

Since $\boldsymbol{x}_1, \boldsymbol{x}_2 \in S$, $(\boldsymbol{x}_1, f(\boldsymbol{x}_1)), (\boldsymbol{x}_2, f(\boldsymbol{x}_2)) \in \text{hyp}(f)$. Because $\text{hyp}(f)$ is convex,

$$\begin{aligned}
&\lambda(\boldsymbol{x}_1, f(\boldsymbol{x}_1)) + (1-\lambda)(\boldsymbol{x}_2, f(\boldsymbol{x}_2)) \\
&\qquad = (\lambda \boldsymbol{x}_1 + (1-\lambda)\boldsymbol{x}_2, \lambda f(\boldsymbol{x}_1) + (1-\lambda) f(\boldsymbol{x}_2)) \in \text{hyp}(f)
\end{aligned}$$

Thus by the definition of a hypograph:

$$\lambda f(\boldsymbol{x}_1) + (1-\lambda) f(\boldsymbol{x}_2) \leq f(\lambda \boldsymbol{x}_1 + (1-\lambda)\boldsymbol{x}_2)$$

Therefore, $f$ is concave.

Example: $S = \left\{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 \leq 1, z = 1\right\}$ is convex, because

- $f(x, y) = x^2 + y^2$ is convex (will be proved later);
- thus $\text{epi}(f) = \left\{(x, y, z) \in \mathbb{R}^3 : z \geq x^2 + y^2\right\}$ is convex;
- $S =$ intersection of $\text{epi}(f)$ with the plane described by $z = 1$, which is also convex

Exercise: Prove that the set $T = \left\{(x, y) : y \geq x^2\right\}$ is convex.

We know that $g(x) = x^2$ is a convex function. Since $T = \text{epi}(g)$, by the previous proposition, $T$ is a convex set.

# First-order conditions for convexity/concavity

## Theorem (compare SB Thm 21.2, 21.3)

*If $S \subseteq \mathbb{R}^n$ is open and convex and $f : S \to \mathbb{R}$ continuously differentiable, then $f$ is*

- *concave if and only if*

$$\forall \boldsymbol{x}, \boldsymbol{x}_0 \in S : \quad f(\boldsymbol{x}) - f(\boldsymbol{x}_0) \leq \nabla f(\boldsymbol{x}_0) \cdot (\boldsymbol{x} - \boldsymbol{x}_0) \tag{6}$$

- *strictly concave if and only if inequality (6) is strict for all $\boldsymbol{x} \neq \boldsymbol{x}_0$.*
- *convex if and only if*

$$\forall \boldsymbol{x}, \boldsymbol{x}_0 \in S : \quad f(\boldsymbol{x}) - f(\boldsymbol{x}_0) \geq \nabla f(\boldsymbol{x}_0) \cdot (\boldsymbol{x} - \boldsymbol{x}_0) \tag{7}$$

- *strictly convex if and only if inequality (7) is strict for all $\boldsymbol{x} \neq \boldsymbol{x}_0$.*

By the theorem, the continuously differentiable function

- $f : \mathbb{R} \to \mathbb{R}$ is concave if and only if for all $x, x_0 \in \mathbb{R}$

$$f(x) - f(x_0) \leq f'(x_0)(x - x_0)$$

or

$$f(x) \leq f(x_0) + f'(x_0)(x - x_0).$$

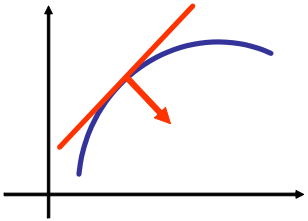
Thus $f$ is concave if and only if for every $x_0 \in \mathbb{R}$, the graph of $f$ lies below the tangent line of $f$ at point $x_0$.

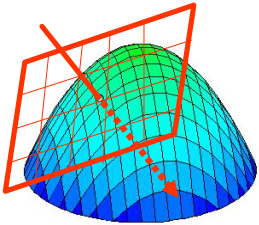
- $f : \mathbb{R}^2 \to \mathbb{R}$ is concave if and only if for all $\boldsymbol{x}, \boldsymbol{x}_0 \in \mathbb{R}^2$

$$f(\boldsymbol{x}) \leq f(\boldsymbol{x}_0) + \nabla f(\boldsymbol{x}_0)(\boldsymbol{x} - \boldsymbol{x}_0).$$

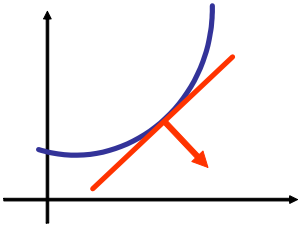
Thus $f$ is concave if and only if for every $\boldsymbol{x}_0 \in \mathbb{R}^2$, the graph of $f$ lies below the tangent plane of $f$ at point $\boldsymbol{x}_0$.

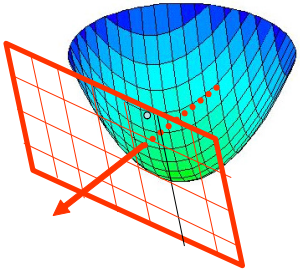
Geometric interpretation of concavity for functions of one or two variables:





Geometric interpretation of convexity for functions of one or two variables:





Example: Consider $f : \mathbb{R}^2 \to \mathbb{R}, (x, y) \mapsto x^2 + y^2$.

- $f$'s tangent plane at point $(x_0, y_0)$ is given by

$$\begin{aligned} g(x, y) &= f(x_0, y_0) + \nabla f(x_0, y_0) \cdot (x - x_0, y - y_0) \\ &= f(x_0, y_0) + \frac{\partial f(x_0, y_0)}{\partial x}(x - x_0) + \frac{\partial f(x_0, y_0)}{\partial y}(y - y_0) \\ &= x_0^2 + y_0^2 + 2x_0(x - x_0) + 2y_0(y - y_0) \\ &= 2x_0 x + 2y_0 y - x_0^2 - y_0^2 \end{aligned}$$

- Thus we obtain

$$f(x, y) - g(x, y) = x^2 + y^2 - (2x_0 x + 2y_0 y - x_0^2 - y_0^2) = (x - x_0)^2 + (y - y_0)^2$$

- As $f(x, y) - g(x, y) > 0$ for all $(x, y) \neq (x_0, y_0)$, we conclude by the previous theorem that $f$ is strictly convex.